Activity 3 Report: Cross-Group Deployment and Verification

COMP 018 Database Administration | Project tested: Binary Sisters, Restaurant Online Ordering & Table Reservation System | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	Group 10 (G10), Binary Sisters, Restaurant Online Ordering & Table Reservation System
Repository and branch tested	https://github.com/kathlenmaecallos11-arch/Restaurant-Online-Ordering-Table-Reservation-System
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a3_verify_privs.txt, a3_tests.txt, screenshots of the setup output and the application

Only the repository and its documentation were used. The repository was not modified.

2. Test environment

Item	Value
Operating system	Windows 11
PostgreSQL	PostgreSQL 18.6 (x86_64 Windows) with psql on PATH. The setup output shows pg_maintain was granted, which the project only does on PostgreSQL 17 or newer
PHP	8.2.12 with pdo_pgsql and pgsql enabled

Item	Value
Database	grp_binarysisters_projectdb, kept separate from other projects
Documents reviewed	README.md, docs/RUN_LOCALLY.md, docs/user-roles.md, docs/user-role-setup.md, docs/database-documentation.md, docs/testing-guide.md, docs/troubleshooting.md

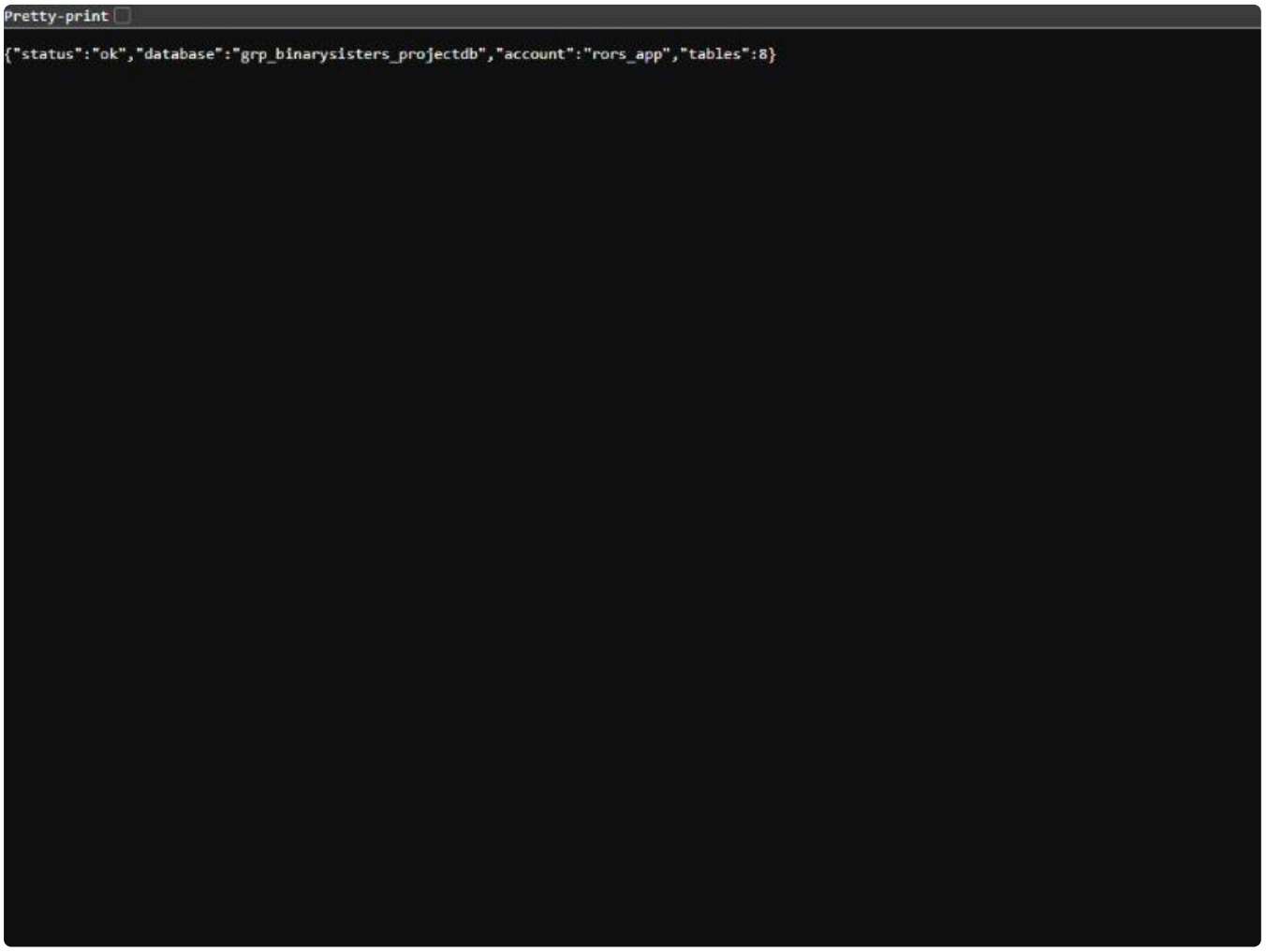
3. Installation procedure and result

Step	Action	Result
1	Extracted the project and opened PowerShell in the project folder	OK
2	Ran .\database\setup.ps1 -WithRoles (scripts were blocked by the PowerShell execution policy, so Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass was used for that session)	Database, tables, constraints, views, indexes and seed data created
3	Role scripts 01_create_roles.sql and 02_grants.sql ran	OK
4	Password step: database/roles/03_set_passwords.psql	**Failed. The file is missing from the repository**
5	Setup continued, ran 04_verify_privileges.sql and printed that the database is ready	26 of 26 PASS
6	Set a password for each rors_ login manually with ALTER ROLE	OK
7	Copied .env.example to .env with DB_USER=rors_app and the rors_app password	OK
8	Enabled the pdo_pgsql and pgsql extensions in php.ini (documented in the project's troubleshooting guide)	OK
9	php tools\test-connection.php	OK Connected as rors_app (file a3_tests.txt)

Step	Action	Result
10	psql -f tests\crud_tests.sql	25 PASS, 0 failed, run as postgres inside a transaction that ended in ROLLBACK, so row counts afterwards were unchanged (62 orders, 171 order lines, 58 payments, 53 reservations). Run as rors_app it stopped at line 16 with: permission denied to create temporary tables in database
11	php -S localhost:8000 -t app/public and http://localhost:8000/health	OK, the health endpoint returned status ok (see the health figure)

Overall setup result: Success with issues. A reader following the README cannot complete the role setup without the missing file.

Figure 1. Application health check at http://localhost:8000/health, returning status ok, database grp_binarysisters_projectdb, account rors_app and 8 tables.



4. Database verification

Seed data loaded (setup output):

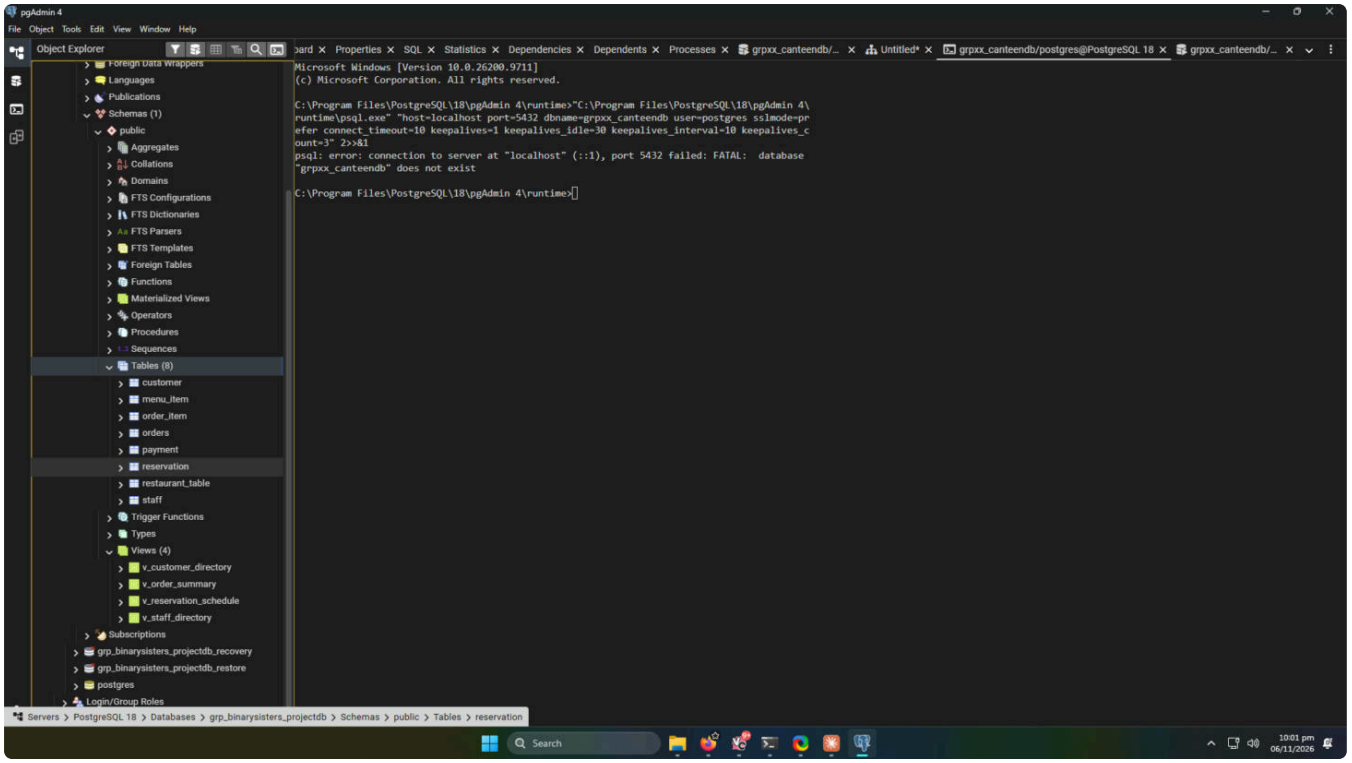
customers	staff	menu items	tables	orders	order items	payments	reservations
51	10	50	20	60	167	56	50

Objects found with \dt+, \dv, \ds and \dx:

- 8 tables: customer, menu_item, order_item, orders, payment, reservation, restaurant_table, staff.
- 4 views: v_customer_directory, v_order_summary, v_reservation_schedule, v_staff_directory.
- 7 sequences, one per SERIAL column.
- Extensions: btree_gist 1.8, pgcrypto 1.4, plpgsql.
- The migrations define 6 foreign keys, 16 CHECK constraints, a double-booking exclusion constraint (ex_reservation_no_overlap) on reservation, and 4 triggers (trg_order_item_keep_one, trg_order_item_total, trg_orders_require_items, trg_reservation_capacity) that keep orders.total_amount in sync and require at least one order item. Activity 6 confirmed that all of these survive a restore.

The project's own privilege check (04_verify_privileges.sql) passed 26 of 26 tests. No rors_ role is a superuser or has CREATEDB or CREATEROLE, and no privileges remain for PUBLIC on the tables.

Figure 2. pgAdmin Object Explorer for grp_binarysisters_projectdb, showing the 8 tables and 4 views in schema public.



5. Application roles checked

Application role	Login used	Actions exercised	Result
Customer	maria.santos@rors.example	Browse menu, add 2 dishes, place an order, reserve a table	Pass. Added Calamares and Cheesy Dynamite and placed order #65 (PHP 390.00, Cash, Pending). It appears under My orders. Reserved Table 1 for 2 guests on 11 Nov 2026 at 10:00 AM (Confirmed). Once reserved, the table no longer appears in the free-table list (figures 5 to 10)

Application role	Login used	Actions exercised	Result
Staff	staff@rors.example	Change an order status, record a payment, open /admin/staff (should be denied)	Pass. An order status was changed (order #64 shows Confirmed) and a payment was recorded. The Orders and Payments pages loaded with data. /admin/staff showed Access denied (figures 11 to 13)
Administrator	admin@rors.example	Create, edit or delete a menu item, open the table form, open /admin/staff	Pass. Menu item changes were made (item 52, test2), and the menu item editor, the New table form and the Staff accounts list all loaded (figures 14 to 16)

The application ran with the least-privilege account rors_app, not a superuser. None of the captured screens showed a database permission denied message. The only denial was the designed Access denied page for Staff on /admin/staff. The second reservation attempt on the same table and time could not be made through the interface, because reserved tables are removed from the availability list. The database-level protection (exclusion constraint ex_reservation_no_overlap) is separate from the interface and was confirmed present after restore in Activity 6.

Figure 3. The running application home page (Mesa, order and reserve) at http://localhost:8000/.

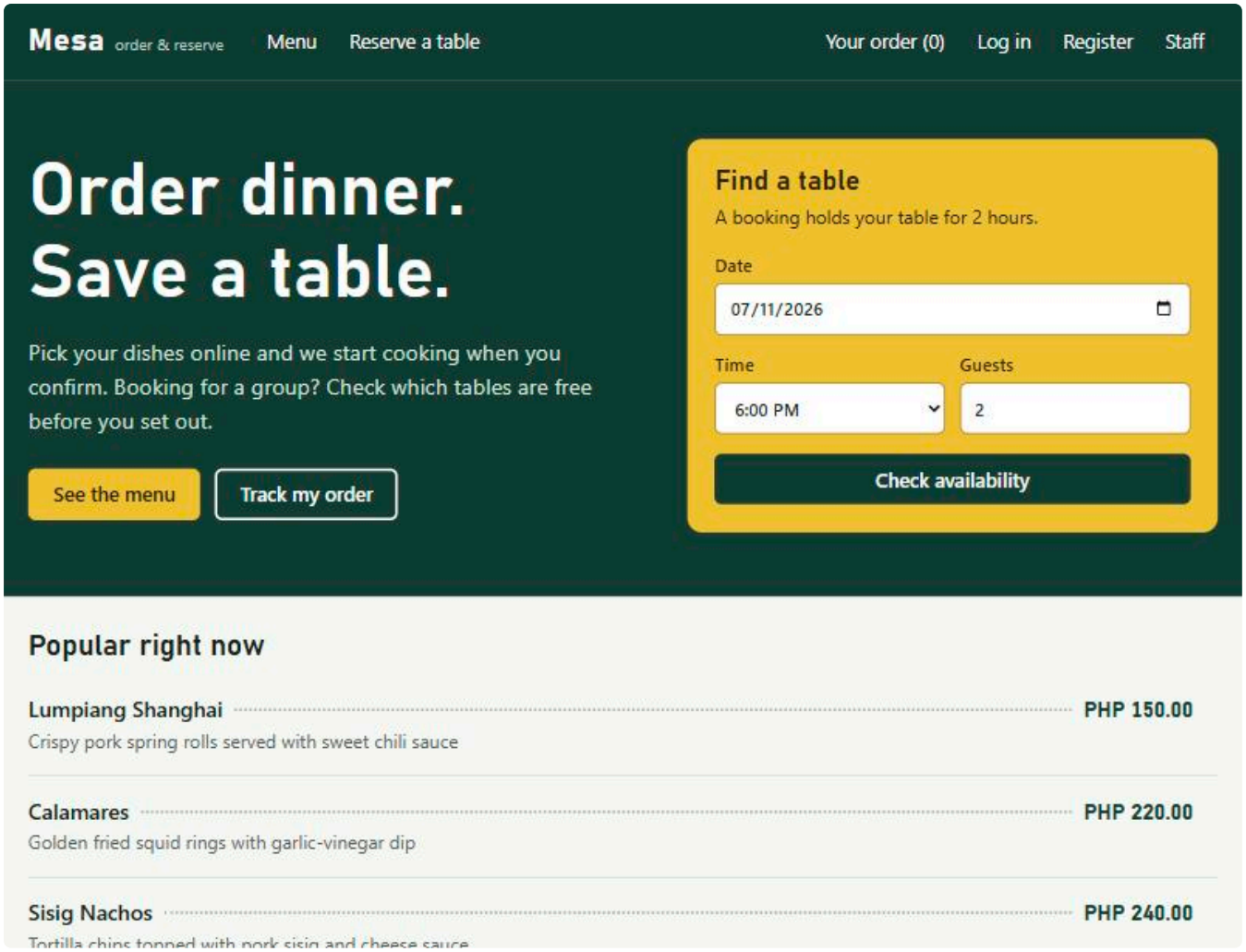
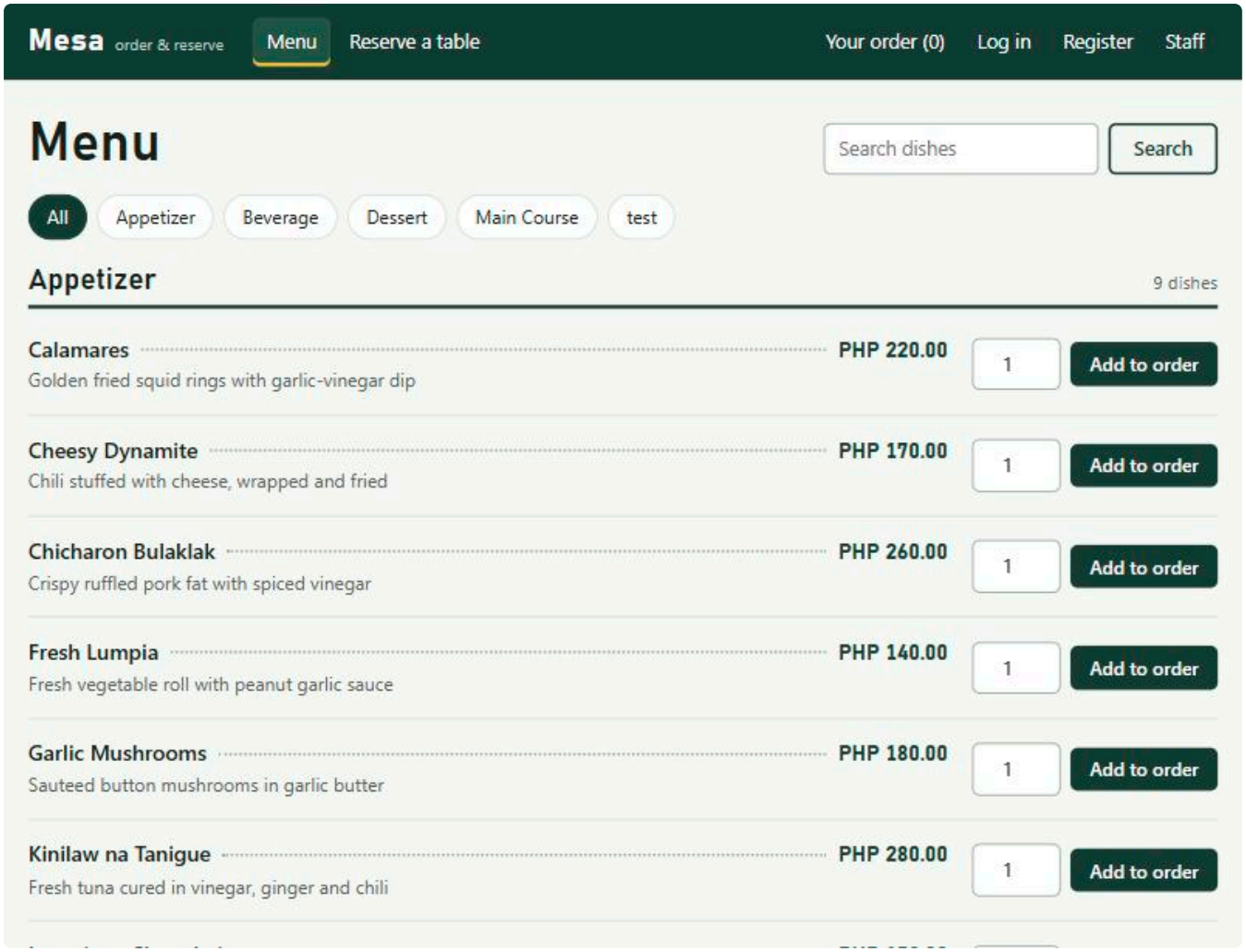


Figure 4. The menu page loaded from the database through the rors_app account.



Customer workflow (rors_app connection)

Figure 5. Cart with two dishes, total PHP 390.00.

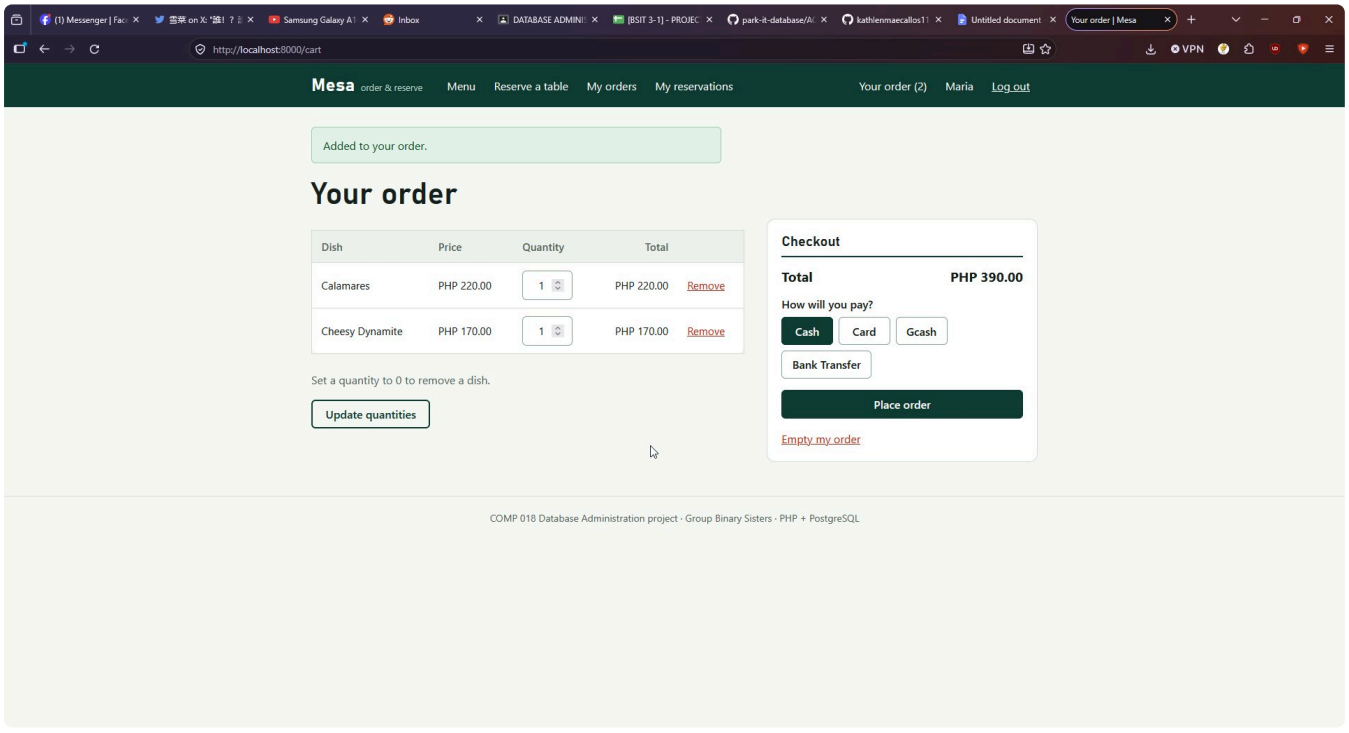


Figure 6. Order #65 placed, status Pending.

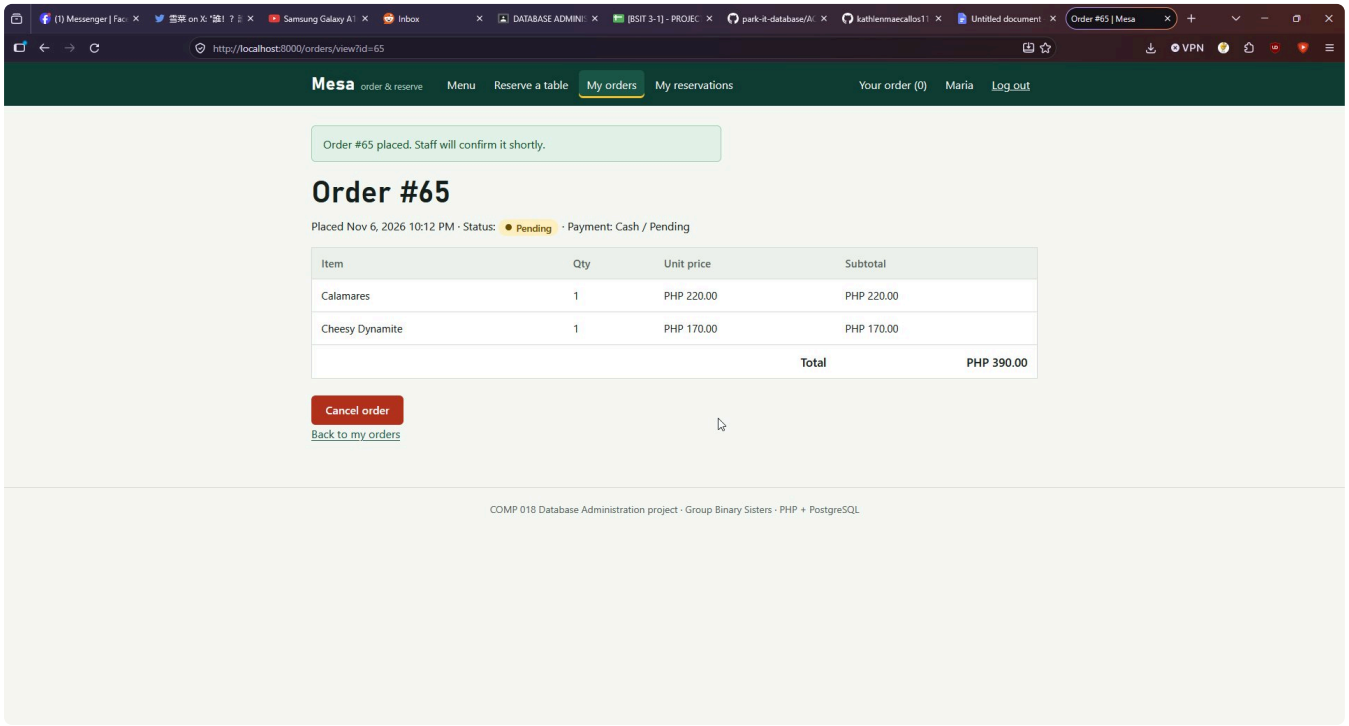


Figure 7. My orders listing order #65.

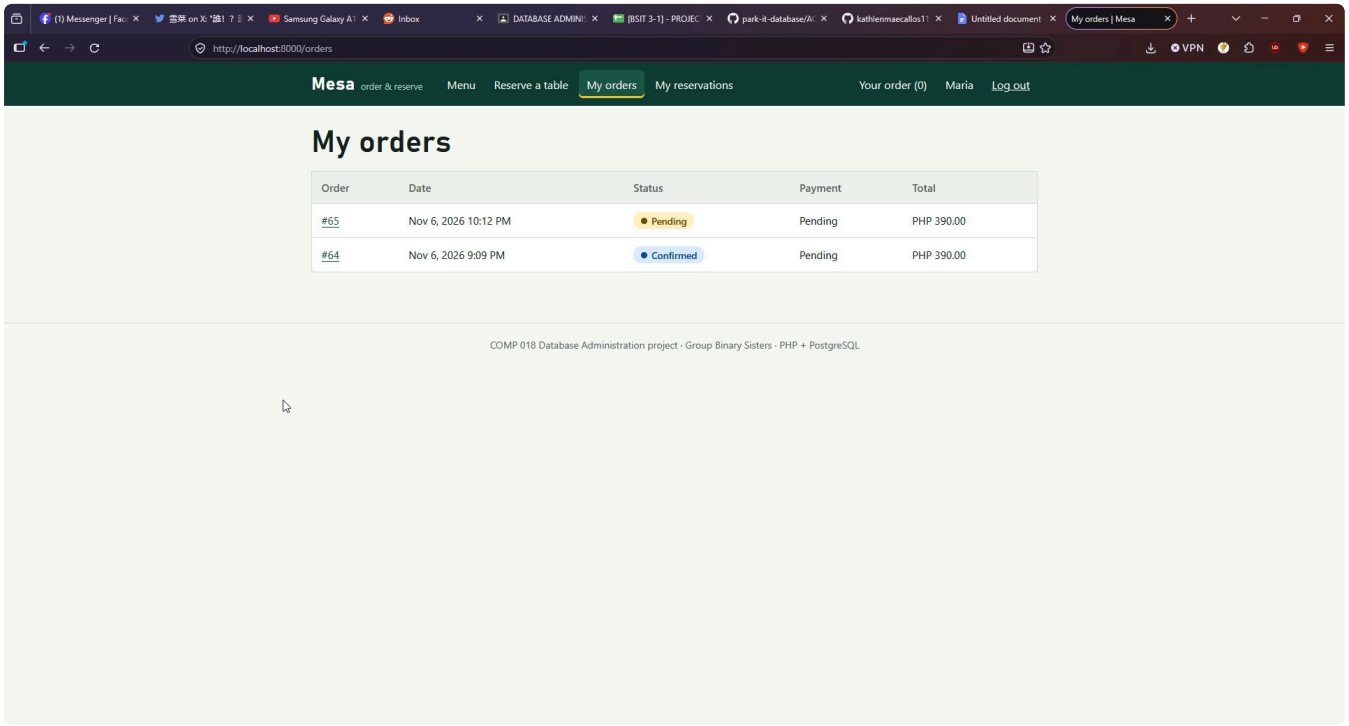


Figure 8. Reservation form with the free tables for 11 Nov 2026 at 10:00 AM, before booking.

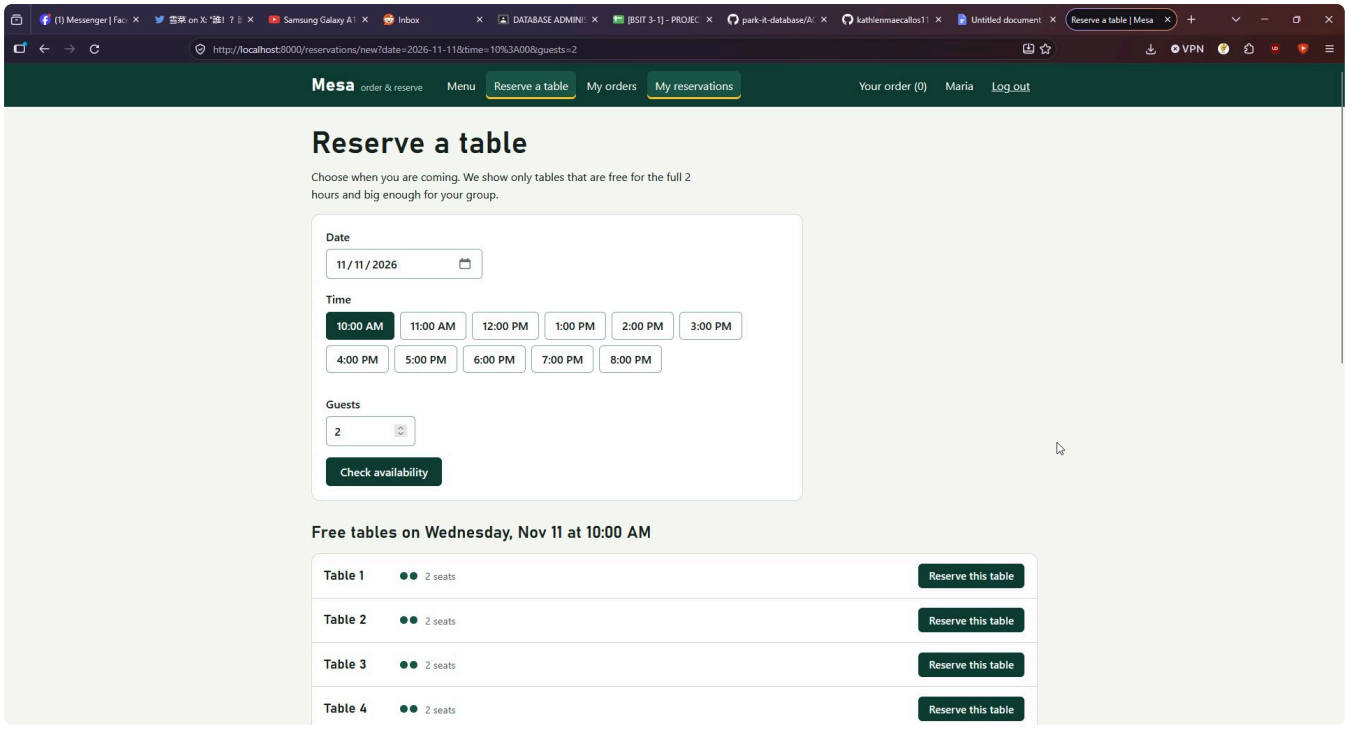


Figure 9. Reservation confirmed for Table 1.

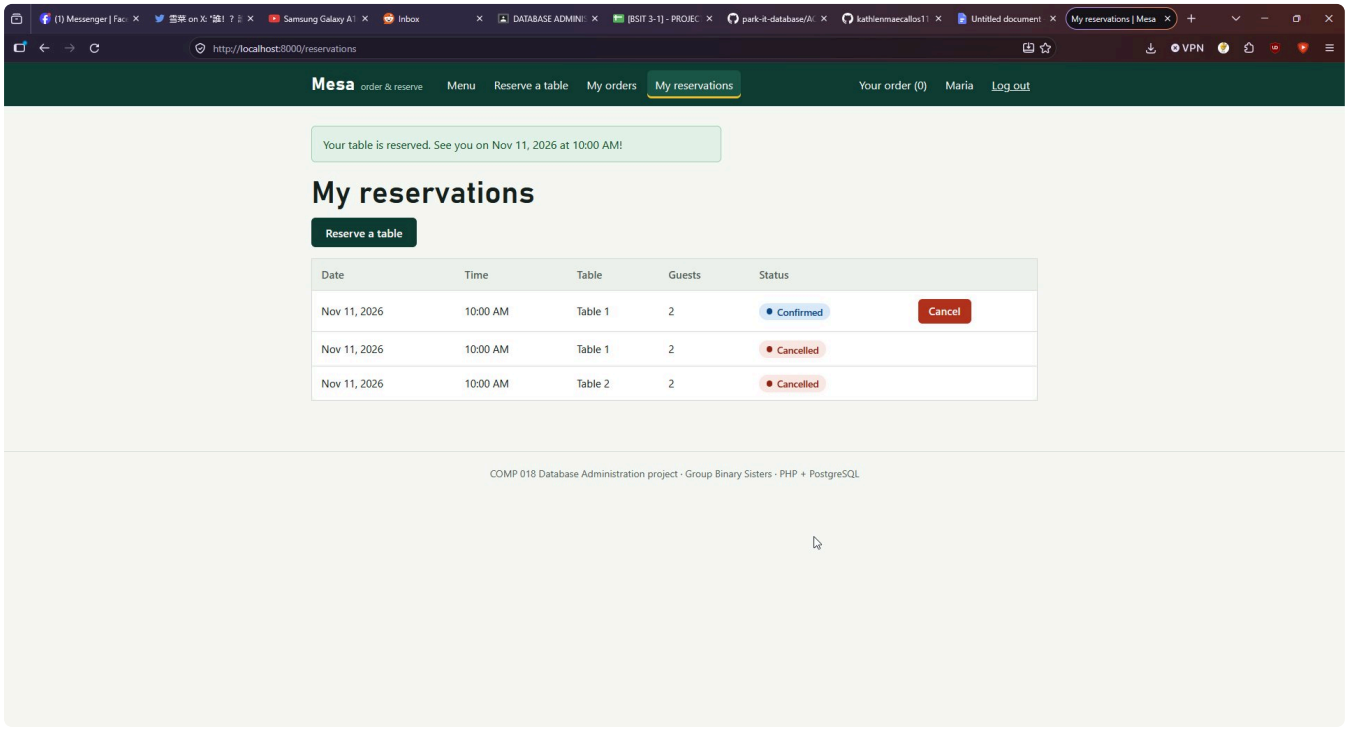
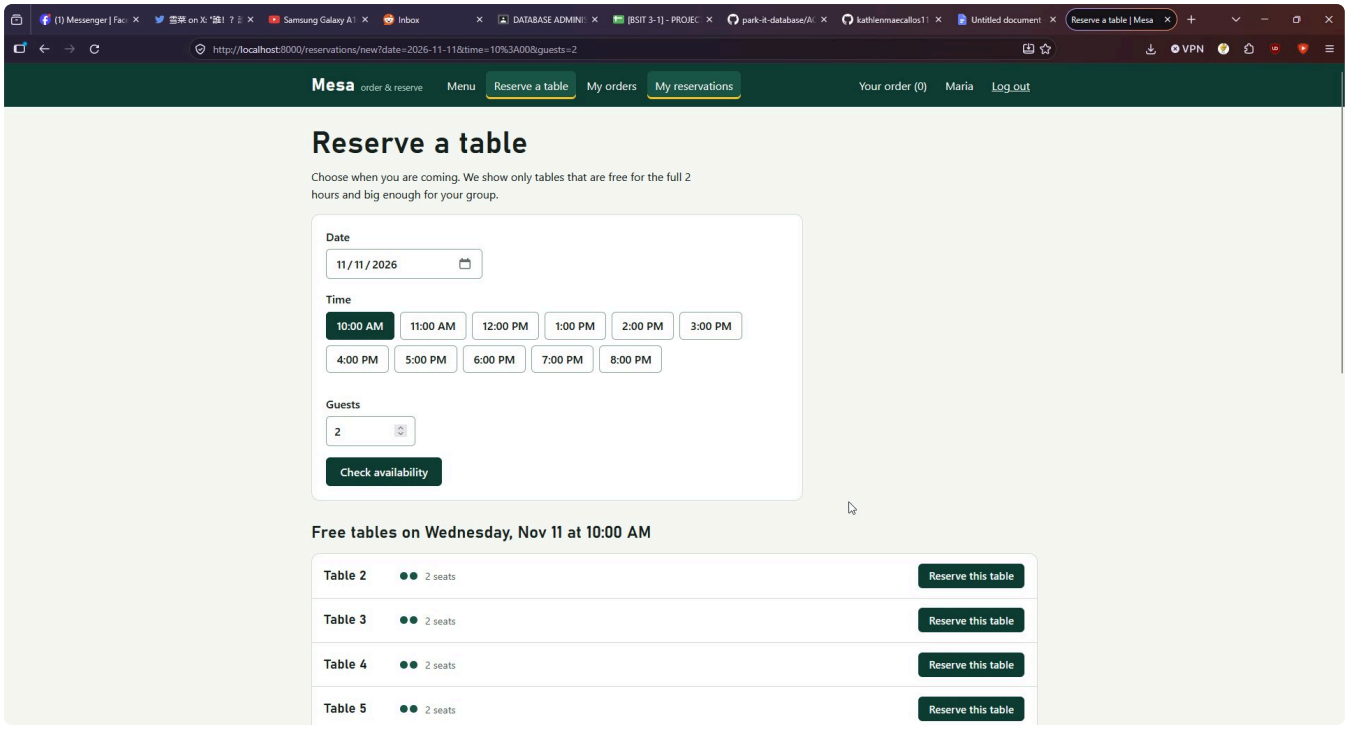


Figure 10. The same date and time afterwards. Table 1 is no longer offered.



Staff workflow (staff@rors.example)

Figure 11. Orders management page.

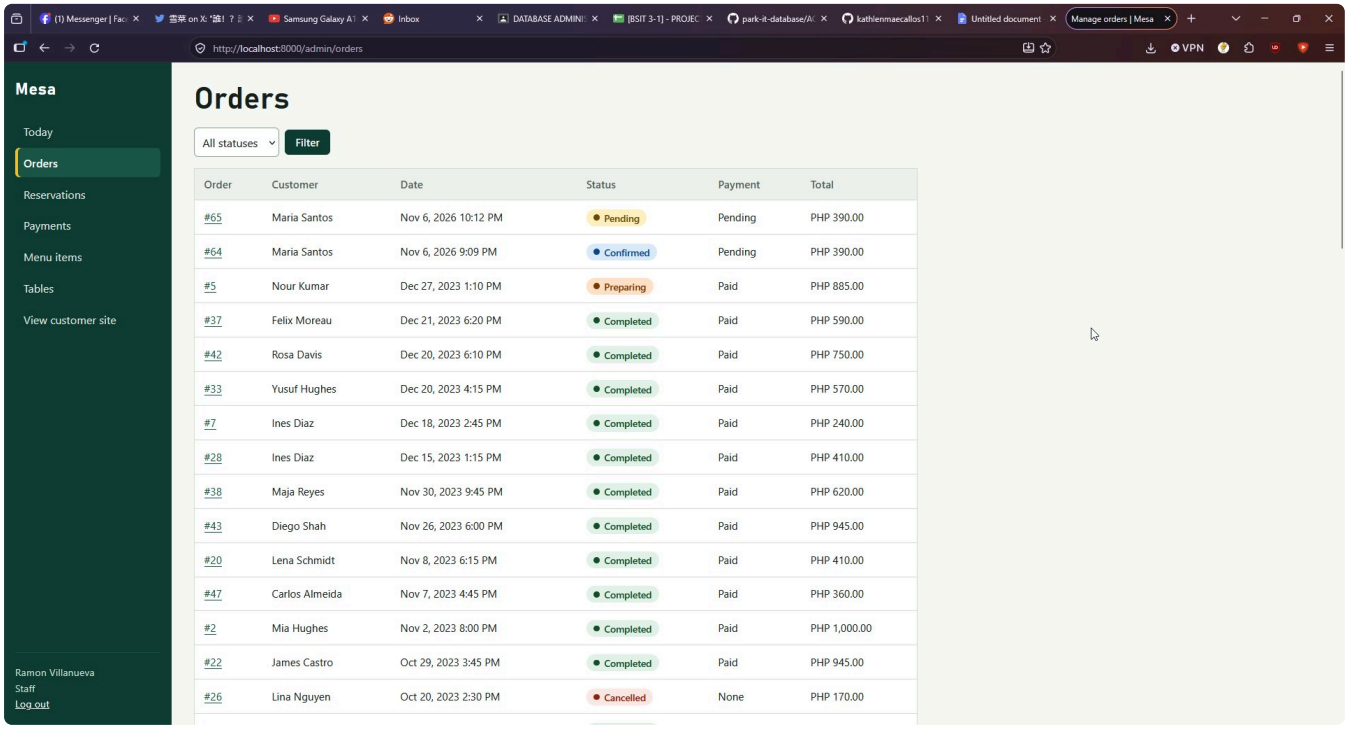


Figure 12. Payments page.

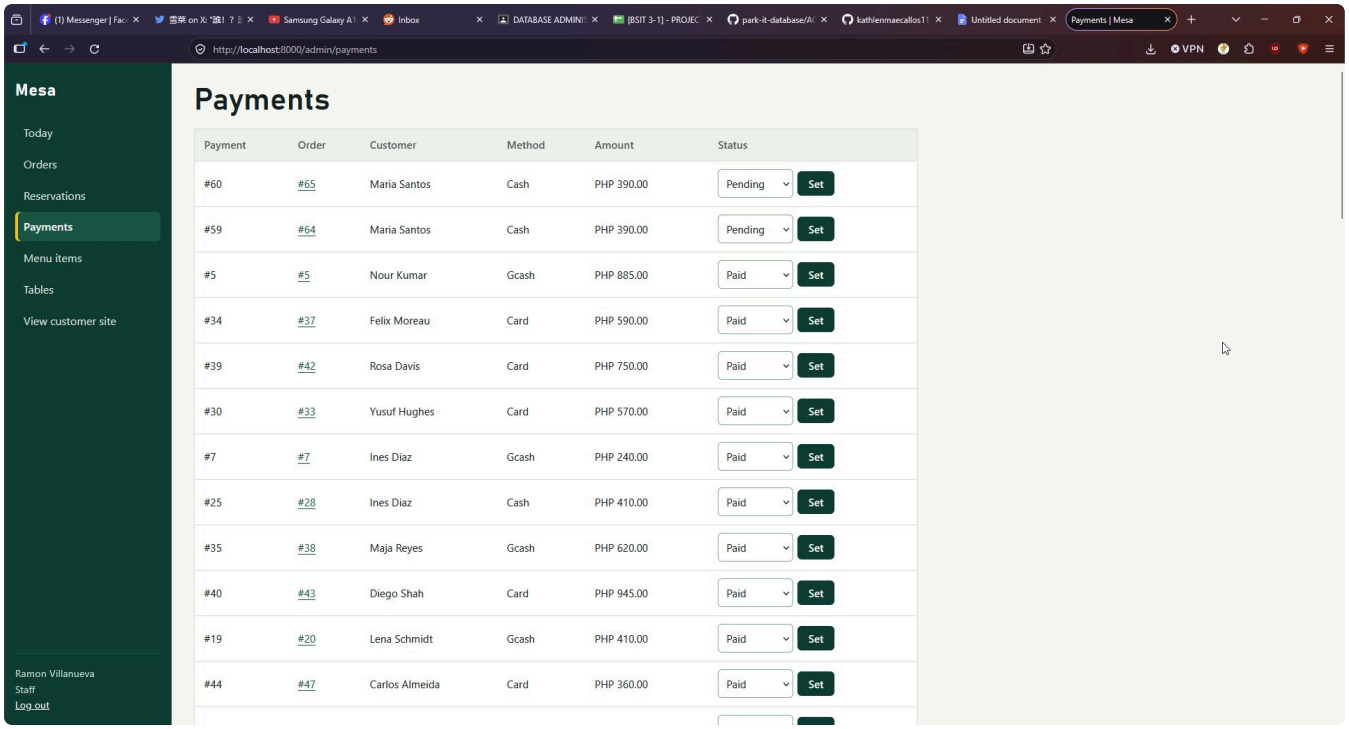
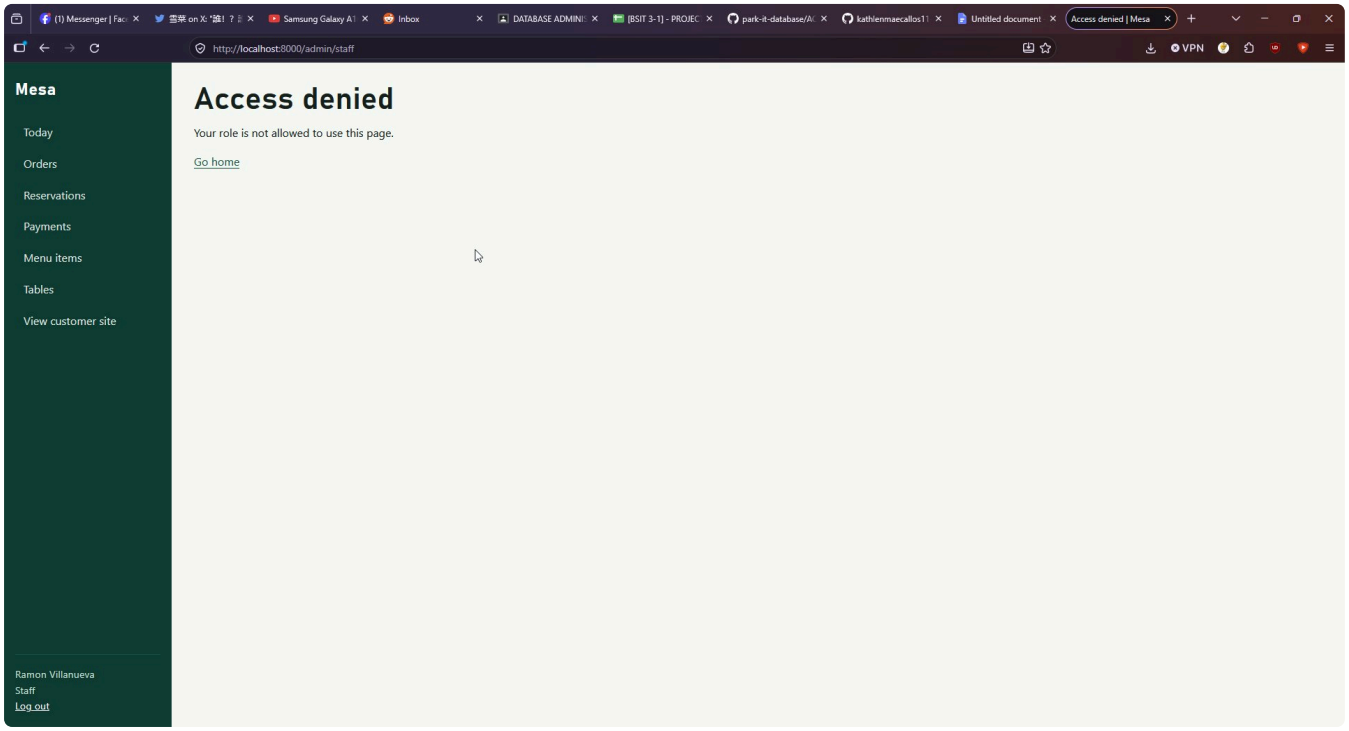


Figure 13. /admin/staff denied for the Staff role.



Administrator workflow (admin@rors.example)

Figure 14. Menu item editor.

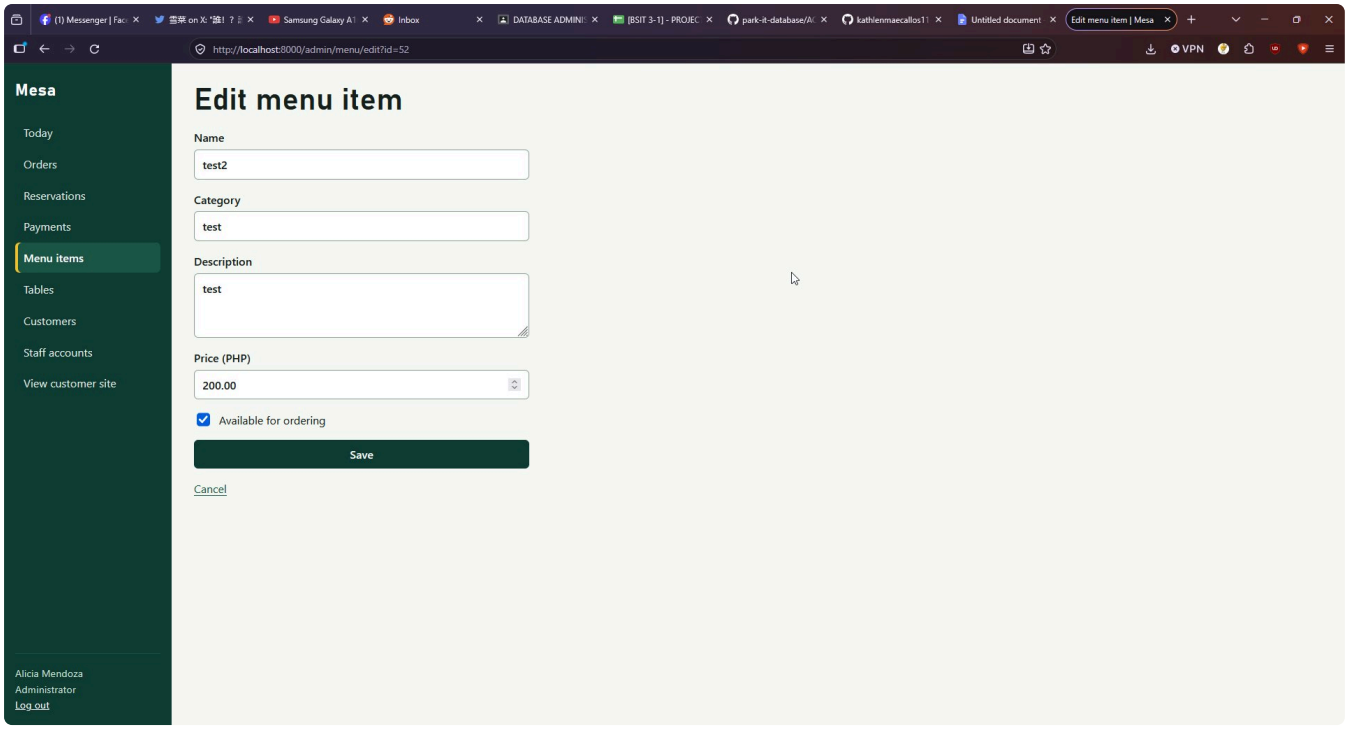


Figure 15. New table form.

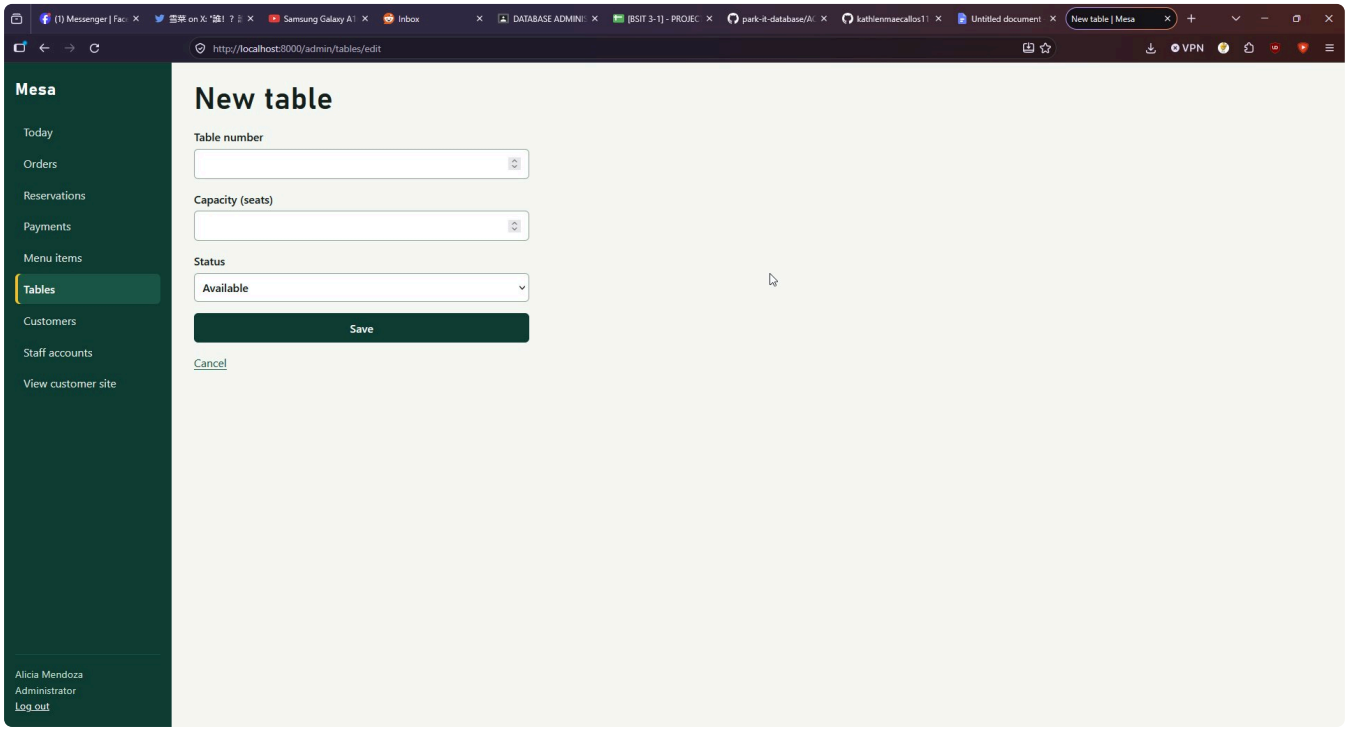
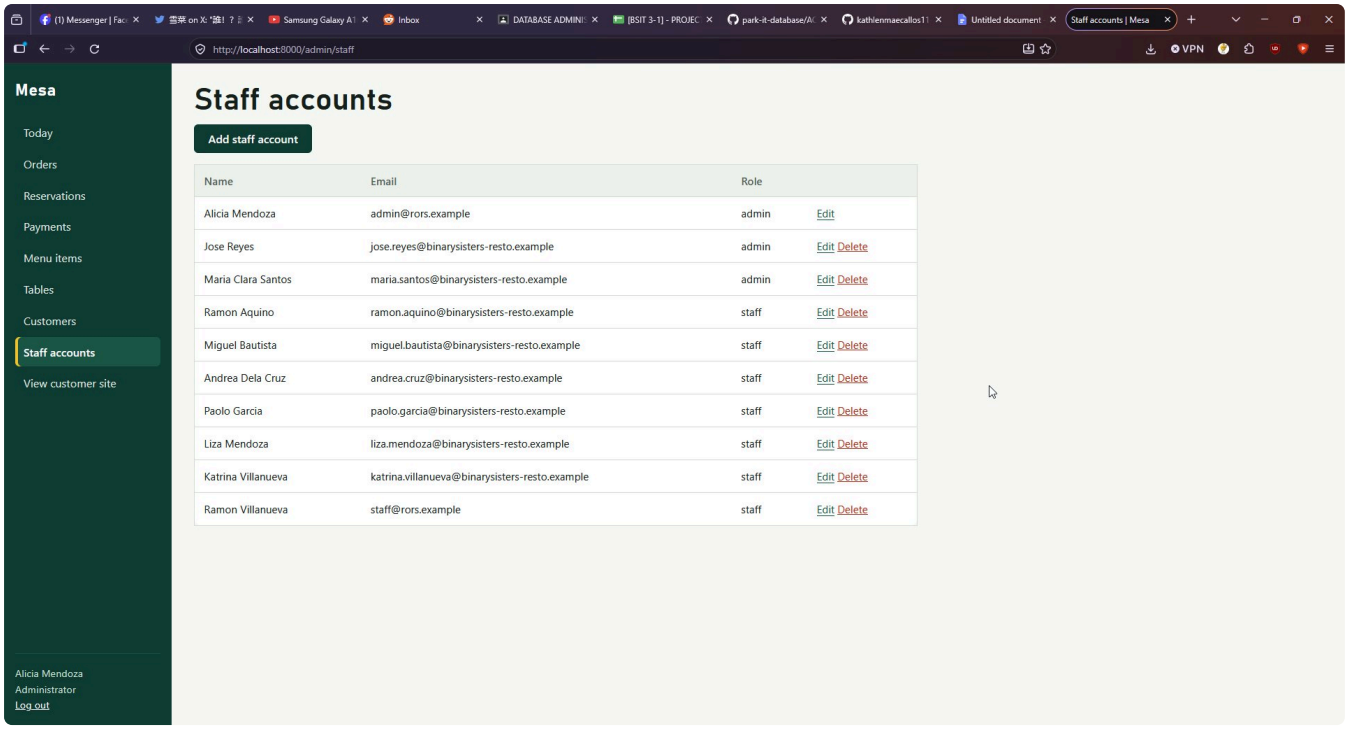


Figure 16. Staff accounts list, available to the Administrator only.



6. Issues found

#	Issue	Type	Impact	Suggested fix
1	database/roles/03_set_passwords.psql is missing, although setup.ps1, setup.sh, the README and the troubleshooting guide all refer to it	Missing instruction or file	All five login accounts are created without a password, so rors_app cannot connect and the application cannot start	Add the file to the repository
2	setup.ps1 continues after the failed password step and ends with a message that the database is ready	Script behaviour	The error is easy to miss and the instructions look as if they succeeded	Stop the script on a failed step, or check the exit code
3	The pdo_pgsql PHP extension was not enabled in our PHP installation	Our environment	None for the owners. The project documents this in the README prerequisites and troubleshooting guide	None needed

7. Feedback sent to owners

Sent to Binary Sisters on 6 November 2026: issues 1 and 2 with the exact error text. A further note: tests/crud_tests.sql fails under the documented runtime account rors_app, which has no TEMP privilege, so the testing guide should say to run it as the administrator. The README, user roles documentation and troubleshooting guide were otherwise clear and complete.

8. Reproducibility conclusion

The project can be reproduced on another Windows machine. The database objects and sample data load correctly and the role verification passes. One undocumented manual step (setting the passwords) is needed because of the missing file.

9. AI usage declaration

- **Tool:** Claude (Anthropic).
- **Tasks:** interpreting the results of the testing.
- **Validation:** every command was run by the group against its own PostgreSQL server, and the results above come from that output.
- **Correction made:** the AI first expected the setup script to stop at the missing file. The real output showed it continued, so that behaviour is recorded as issue 2.